



Task - 03

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Submitted to:

Dr. Raihan Ul Islam, Associate Professor, Department of Computer Science and Engineering

Submitted By

NAME	ID
Fariya Ahmed	2022-2-50-009
Jubaiya Akter Shibly	2022-3-60-255
Nipa Mridha	2022-3-60-254
Sumaiya Akter	2022-1-60-300
Abdullah Al Mahfuz	2020-2-60-066

GROUP : 08

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Abstract

Cracks in concrete structures are a common issue that can affect the safety and lifespan of buildings. In this work, a custom Convolutional Neural Network (CNN) model is developed to classify crack and non-crack images using the CrackDB dataset. The proposed model consists of multiple convolutional and pooling layers followed by fully connected layers to extract discriminative features. The model was trained for 20 epochs, achieving a training accuracy of 95.94% and a validation accuracy of 95.83%. Experimental results show that the proposed CNN achieves a test accuracy of 96.88%, demonstrating its effectiveness in detecting cracks from concrete surface images. The results indicate that a well-designed custom CNN can perform efficiently without relying on complex pretrained architectures.

Introduction

Among various types of structural damage, concrete cracks are one of the most common and critical issues. Traditional manual inspection methods are time-consuming and costly.

With recent advances in deep learning, Convolutional Neural Networks (CNNs) have shown strong performance in image-based classification tasks. While pretrained models such as VGG and ResNet are effective, they often require higher computational resources. Therefore, this work focuses on designing a **custom CNN model** that achieves high accuracy with lower complexity.

The objective of this task is to:

- Design a custom CNN architecture
- Train and evaluate the model on crack images
- Analyze accuracy and classification performance

Methodology

This section describes the dataset preparation, preprocessing techniques, custom CNN architecture and training strategy used for concrete crack classification.

A. Dataset Preparation

The CrackDB dataset was used in this task for binary image classification. The dataset contains labeled concrete surface images belonging to two classes: with crack and without crack. The dataset was divided into training, validation and test sets to ensure unbiased performance evaluation. The training set was used for model learning, the validation set for hyperparameter tuning and the test set for final performance assessment.

B. Image Preprocessing and Augmentation

Before training, all images were resized so that all inputs had the same size across the dataset. Pixel values were normalized to the range $[0,1]$ to improve convergence during training. To enhance model generalization and reduce overfitting, data augmentation techniques were applied, including random rotation, horizontal flipping and zooming. These transformations increase dataset variability while preserving class labels.

C. Custom CNN Model

A custom Convolutional Neural Network (CNN) was designed for binary image classification. The model consists of convolutional layers followed by max-pooling layers for feature extraction. ReLU activation functions were used in hidden layers and a sigmoid activation function was used in the output layer to classify images as crack or non-crack.

D. Training Strategy and Configuration

The CNN model was trained using the Adam optimizer with binary cross-entropy as the loss function. Training was conducted for 20 epochs with a batch size of 32. During training, both training and validation metrics were monitored to evaluate convergence behavior and prevent overfitting.

Parameter	Value
Model Type	Custom CNN
Optimizer	Adam
Epochs	20

Loss Function	Binary Crossentropy
Batch size	32
Activation Function	ReLU, Sigmoid
Dataset	CrackDB

Table1: summarizes the training configuration used in this study.

Results and Discussion

A. Training Performance

The model was trained for 20 epochs. Both training and validation accuracy increased steadily, while loss values decreased, indicating stable learning.

- **Final Training Accuracy:** ~95.94%
- **Final Validation Accuracy:** ~95.83%

B. Test Accuracy

After training, the model was evaluated on the test dataset.

Test Accuracy: 96.88%

This confirms that the model generalizes well to unseen data.

C. Classification Report

The classification performance is summarized in Table 2.

Class	Precision	Recall	F1-score
With crack	0.97	0.97	0.97
Without crack	0.94	0.94	0.94
Overall Accuracy	-	-	0.96

D. Confusion Matrix Analysis

The confusion matrix obtained from the test set is shown below:

$$\begin{bmatrix} 59 & 2 \\ 2 & 33 \end{bmatrix}$$

Out of 96 test samples, the model correctly classified 92 images, misclassifying only 4 samples. The low number of false positives and false negatives demonstrates the robustness of the custom CNN model in crack detection tasks.

Discussion-

The custom CNN achieved high accuracy despite having a simpler architecture compared to pretrained networks. The model successfully learned discriminative features of cracks and showed balanced precision and recall for both classes. This demonstrates that a well-designed custom CNN can be an effective and efficient solution for crack detection tasks.

Conclusion

This study presented a custom CNN-based approach for concrete crack detection using image classification. The model achieved a test accuracy of 96.88%, along with strong precision, recall and F1-scores. Due to its lower computational complexity and high performance, the proposed method is suitable for real-world structural health monitoring applications.

Future work may include testing on larger datasets and deploying the model in real-time inspection systems.